

Q1 JEE Main 2020 - 8 Jan (Morning)

Let $f(x) = (\sin(\tan^{-1}x) + \sin(\cot^{-1}x))^2 - 1$, $|x| > 1$. If $\frac{dy}{dx} = \frac{1}{2} \frac{d}{dx}(\sin^{-1}(f(x)))$ and $y(\sqrt{3}) = \frac{\pi}{6}$, then $y(-\sqrt{3})$ is equal to :

- (A) $\frac{5\pi}{6}$
- (B) $-\frac{\pi}{6}$
- (C) $\frac{\pi}{3}$
- (D) $\frac{2\pi}{3}$

Q2 JEE Main 2020 - 2 Sep (Evening)

If $y = \sum_{k=1}^6 k \cos^{-1}\left\{\frac{3}{5} \cos kx - \frac{4}{5} \sin kx\right\}$, then $\frac{dy}{dx}$ at $x = 0$ is

Q3 JEE Main 2020 - 3 Sep (Morning)

$2\pi - \left(\sin^{-1} \frac{4}{5} + \sin^{-1} \frac{5}{13} + \sin^{-1} \frac{16}{65}\right)$ is equal to:

- (A) $\frac{\pi}{2}$
- (B) $\frac{7\pi}{4}$
- (C) $\frac{3\pi}{2}$
- (D) $\frac{5\pi}{4}$

Q4 JEE Main 2020 - 5 Sep (Morning)

If S is the sum of the first 10 terms of the series $\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{13}\right) + \tan^{-1}\left(\frac{1}{21}\right) + \dots$ then $\tan(S)$ is equal to :

- (A) $-\frac{6}{5}$
- (B) $\frac{5}{11}$
- (C) $\frac{11}{10}$
- (D) $\frac{5}{6}$

Answer Key

Q1 (A) Q2 (91) Q3 (C) Q4 (D)

Q1

Let $\tan^{-1} x = \theta$

$$\Rightarrow x = \tan \theta \Rightarrow \sin \theta = \frac{x}{\sqrt{1+x^2}}$$

$$y = \left(\frac{x}{\sqrt{1+x^2}} + \frac{1}{\sqrt{1+x^2}} \right)^2 - 1$$

$$\Rightarrow y = \frac{(x+1)^2}{1+x^2} - 1$$

$$\Rightarrow y = \frac{2x}{1+x^2} = f(x)$$

Now,

$$\frac{dy}{dx} = \frac{1}{2} \frac{1}{\sqrt{1-f^2}} \times f'(x) = \frac{1}{2} \frac{1}{\sqrt{1-\frac{4x^2}{(1+x^2)^2}}} f'(x) = \frac{(1+x^2)}{2(x^2-1)} f'(x)$$

$$= \frac{1+x^2}{2(x^2-1)} \times 2 \frac{1+x^2-2x^2}{(1+x^2)^2}$$

$$\int dy = \int \frac{1-x^2}{(x^2-1)(1+x^2)} dx$$

$$y = -\tan^{-1} x + c$$

Given,

$$y(\sqrt{3}) = \frac{\pi}{6} \Rightarrow \frac{\pi}{6} = -\frac{\pi}{3} + c$$

$$\Rightarrow c = \frac{\pi}{2}$$

$$y = -\tan^{-1} x + \frac{\pi}{2} = \cot^{-1} x$$

Now,

$$y(-\sqrt{3}) = \cot^{-1} x(-\sqrt{3}) = \frac{5\pi}{6}$$

Q2

$$y = \sum_{k=1}^6 k \cos^{-1} \left\{ \frac{3}{5} \cos kx - \frac{4}{5} \sin kx \right\}$$

$$\text{Let } \cos \alpha = \frac{3}{5} \text{ and } \sin \alpha = \frac{4}{5}$$

$$y = \sum_{k=1}^6 k \cos^{-1} \{ \cos \alpha \cos kx - \sin \alpha \sin kx \}$$

$$= \sum_{k=1}^6 k \cos^{-1} (\cos(kx + \alpha))$$

$$= \sum_{k=1}^6 k(kx + \alpha) = \sum_{k=1}^6 (k^2 x + \alpha k)$$

$$\frac{dy}{dx} = \sum_{k=1}^6 k^2 = \frac{6(7)(13)}{6} = 91$$

Q3

$$2\pi - \left(\sin^{-1} \frac{4}{5} + \sin^{-1} \frac{5}{13} + \sin^{-1} \frac{16}{65} \right)$$

$$= 2\pi - \left(\tan^{-1} \frac{4}{3} + \tan^{-1} \frac{5}{12} + \tan^{-1} \frac{16}{63} \right)$$

$$= 2\pi - \left\{ \tan^{-1} \left(\frac{\frac{4}{3} + \frac{5}{12}}{1 - \frac{4}{3} \cdot \frac{5}{12}} \right) + \tan^{-1} \frac{16}{63} \right\}$$

$$= 2\pi - \left(\tan^{-1} \frac{63}{16} + \tan^{-1} \frac{16}{63} \right)$$

$$= 2\pi - \left(\tan^{-1} \frac{63}{16} + \cot^{-1} \frac{63}{16} \right)$$

$$= \frac{3\pi}{2}$$

Q4

$$S = \tan^{-1} \left(\frac{1}{3} \right) + \tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{13} + \dots$$

$$= \tan^{-1} \left(\frac{1}{1+1 \cdot 2} \right) + \tan^{-1} \left(\frac{1}{1+2 \cdot 3} \right) + \tan^{-1} \dots$$

$$= \tan^{-1} \left(\frac{2-1}{1+2 \cdot 1} \right) + \tan^{-1} \left(\frac{3-2}{1+3 \cdot 2} \right) + \tan^{-1} \left(\frac{4-3}{1+3 \cdot 4} \right) + \dots + \tan^{-1} \left(\frac{11-10}{1+11 \cdot 10} \right)$$

$$= (\tan^{-1} 2 - \tan^{-1} 1) + (\tan^{-1} 3 - \tan^{-1} 2) +$$

$$= \tan^{-1} 4 - \tan^{-1} 3) + \dots + (\tan^{-1} 11 - \tan^{-1} 10)$$

$$= \tan^{-1} 11 - \tan^{-1} 1$$

$$= \tan^{-1} \left(\frac{11-1}{1+11 \cdot 1} \right)$$

$$\tan(S) = \frac{5}{6}$$